| | | | |
|---|---|---|---|
| **Fiscal Year:** | FY 2005 | **Task Last Updated:** | FY 10/26/2004 |
| **PI Name:** | Ivanchenko, Maria G. Ph.D. | | |
| **Project Title:** | Effect Of Light On Plant Gravitropism: Functional Genomic Analysis | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | SPACE BIOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Organismal biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Plant Biology | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | ivanchem@science.oregonstate.edu | **Fax:** | FY 541-737-3573 |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 541-737-5279 |
| **Organization Name:** | Oregon State University | | |
| **PI Address 1:** | Department Botany and Plant Pathology | | |
| **PI Address 2:** | 2082 Cordley Hall | | |
| **PI Web Page:** | http://osu.orst.edu/ | | |
| **City:** | Corvallis | **State:** | OR |
| **Zip Code:** | 97331 | **Congressional District:** | 5 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2001 Space Biology 01-OBPR-06 |
| **Start Date:** | 11/01/2002 | **End Date:** | 10/31/2006 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 2 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Lomax, Terri L. Ph.D. ( Oregon State University )<br>Fowler, John ( Oregon State University ) | | |
| **Grant/Contract No.:** | NAG2-1593 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

| | |
|---|---|
| **Task Description:** | The response of higher plants to gravity is controlled by integrated interactions among environmental and endogenous signals. Interactions between light and the gravitropic response have long been recognized, but the underlying mechanism is poorly understood. Fundamental knowledge in this field will allow effective application of light to improve plant growth in space, which is essential biological knowledge to support NASA's goals for long-duration human presence and exploration in space. Tomato plants carrying a mutation in the Lazy-2 gene exhibit a reversed gravitropic response under red light and grow downward instead of upward. Further study of the lazy-2 tomato mutant provides a unique opportunity to understand the mechanism that permits light to modulate the gravitational set-point angle of shoots. To our knowledge, the lazy-2 lesion is the only mutation that reverses the directionality of the gravitropic response rather than reducing or eliminating it. Our goal is to utilize the lazy-2 lesion to elucidate the molecular mechanisms underlying light regulation of gravicurvature. To this end, we plan to 1) use auxin-responsive gene constructs as reporters to study changes in auxin transport and sensitivity, 2) identify molecular components from the Lazy-2 signal transduction pathway via global expression profiling, 3) clone the Lazy-2 gene to identify its role in light regulation of the gravitropic response, and 4) perform comparative studies of the phylogeny of the Lazy-2 gene and other genes involved in light/gravity interactions.<br>The proposed research supports the goals of NASA's Fundamental Space Biology Program in determining the role of gravity on normal plant development; developing an understanding at the genetic, molecular, and cellular levels of how the expression of genes is controlled by the force of gravity and light; and establishing how the mechanisms by which plants perceive, transduce, integrate, and respond to gravity and red light compare among plant species. Tomato plants carrying a mutation in the Lazy-2 gene display a striking phenotype. Under red light, lazy-2 mutants exhibit a reversed gravitropic response and grow downward instead of upward. In darkness or blue light, the mutants assume the upright growth habit characteristic of wild-type plants, indicating that the Lazy-2 gene product acts in a branch of the shoot gravitropic pathway that is specifically regulated by red light. Our goal is to utilize the lazy-2 lesion to elucidate the molecular mechanisms underlying light regulation of gravicurvature. The aim of this project is to enhance our understanding of how the combined forces of gravity and light interact to regulate plant development, which will lead to the discovery of mechanisms that insure or refine plant production in space. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Not Available |
| **Task Progress:** | We have screened about 4,000 plants and have identified 8 plants with recombinations close to Lazy-2 on chromosome 5, allowing the placement of new markers close to the Lazy-2 gene on chromosome 5. Marker TG344 was localized only 0.04 cM from Lazy-2. This allowed the assembly of a contiguous architecture of overlapping sequences (contig) of at least 200-300 kb length over the region of chromosome 5 where the lazy-2 gene resides.<br>We have also transformed wild type and lazy-2 plants with reporter constructs that monitor how the plant hormone auxin accumulates upon gravistimulation under different light conditions. We observed that in darkness, both wild-type and lazy-2 hypocotyls express an auxin-response reporter on the lower, expanding side, while under red light, the lazy-2 mutant expresses the auxin reporter on the upper side, suggesting that the direction of lateral auxin transport and tissue responsiveness to auxin are reversed in the mutant.<br><br>We also used auxin-transport inhibitors and chemicals affecting the integrity of the actin cytoskeleton to screen for differences between wild type and lazy-2 gravitropic responses in red light. Our observations indicate that alterations within the actin cytoskeleton might account for the generation of the lazy-2 mutant phenotype. The Lazy-2 gene product might act early in the gravitropic response, likely in the poorly understood part of the pathway between perception of gravity and lateral redistribution of auxin, which involves processes within the actin cytoskeleton. |
| **Bibliography Type:** | Description: (Last Updated: 04/10/2007) |
| **Abstracts for Journals and Proceedings** | Ivanchenko, M.; Jetnick, Z.; Lomax, T. "Cloning and expression of LeIAA3 and LeIAA11 auxin-response proteins from tomato."<br>In the present study, we report cloning and expression of two new full-length Aux/IAA auxin response genes, LeIAA3 and LeIAA11, from tomato and their promoter regions. The two encoded proteins are nuclear localized and show 50-60% homology to known Aux/IAA proteins from Arabidopsis. The promoter regions of LeIAA3 and LeIAA11 are auxin-inducible and confer tissue-specific expression to the GUS reporter. The regions contain several DNA motifs showing similarity with auxin-response elements from the PS IAA4/5 and GH3 promoters.<br><br>in preparation , Jan-2005 |
| **Abstracts for Journals and Proceedings** | Ivanchenko, M.; Mengucci, A.; White, T.; Findlay, K.; Oh, K.; Lomax, T. "The intersection between the red light and gravity response mechanism in plant shoots."<br>The response of higher plants to gravity is regulated by integrated interactions among environmental and endogenous signals. Interactions between light and the gravitropic response have long been recognized, but the underlying mechanism is poorly understood. Fundamental knowledge in this field will allow effective application of light to improve plant growth in space, essential biological knowledge to support NASA's goals for long-duration human presence and exploration of space. The lazy-2 mutant of tomato demonstrates the unique phenomenon of reversed gravicurvature of plant shoots upon exposure to red light. Our objective is to utilize the lazy-2 lesion to elucidate the molecular mechanism of light regulation of the gravitropic response.<br><br>ASGSB Annual Meeting, Alabama, 2003. , Nov-2003 |

| | |
|---|---|
| **Abstracts for Journals and Proceedings** | Findlay, K.; Ivanchenko, M.; Lomax, T. "Red light induces irreversible changes in auxin transport of dark-grown tomato seedlings."<br>We used auxin-transport inhibitors, the DR5 auxin reporter, and chemicals affecting the integrity of the actin cytoskeleton to screen for differences between wild-type and lazy-2 gravitropic responses in red light. We demonstrate that the actin cytoskeleton is involved in the lazy-2 mutant phenotype. Moreover, lateral auxin transport upon gravitstimulation is reversed in lazy-2. The Lazy-2 gene product therefore appears to act early in the gravitropic response, likely in the poorly-understood part of the pathway between the perception of gravity and the lateral redistribution of auxin.<br><br>Auxin'2004 Conference, Colympary, Greece. , May-2004 |
| **Abstracts for Journals and Proceedings** | Ivanchenko, M.; Lomax, T. "Roles of a cyclophilin, LeCYP1/DGT, in auxin-mediated tomato development."<br>The loss-of-function dgt tomato mutant has an auxin-resistant phenotype marked by reduced adult size and dramatic alterations in root development with no lateral root formation. In this aspect, the dgt mutation resembles the gain-of-function mutations in AtIAA14 (SLR1, Fukaki et al., 2002) and AtIAA28 (Rogg et al., 2001) early auxin-response genes of Arabidopsis. We compared the expression of AtIAA14, AtIAA28, and AtIAA2 GUS-tagged promoters in dgt and wild type tomato plants. Our observations indicate that the specificity of expression of all three promoters is compromised in dgt. This indicates that the LeCYP1 PPIase is a major regulator in auxin signaling, acting upstream of multiple auxin-response gene promoters to regulate and coordinate their expression. The LeCYP1 protein localizes to the nucleus and possibly to the Golgi endomembrane system, suggesting that both cellular compartments are likely sites of its action.<br><br>Auxin'2004 Conference, Colympary, Greece. , May-2004 |